

Data Anonymization Based on Natural Equivalent Class

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Abstract—Data anonymization is widely used to preserve the utility of published datasets without compromising privacy. The state-of-the-art data anonymization approaches are mainly single-record-based algorithms. They group similar records together one by one, then form equivalence classes through generalization. However, these algorithms didn't utilize equivalence classes which exist in the raw dataset. In this paper, we propose a new concept named natural equivalent class. It refers to the record set with the same quasi-identifier values naturally existing in the raw dataset. We theoretically prove that the natural equivalent class can effectively reduce the computational complexity of clustering algorithms as well as information loss. Then, we propose a novel clustering-based anonymization algorithm, which tries to cluster records without separating any natural equivalent class. Extensive experiments on real world datasets show that our approach outperforms the previous clustering-based anonymization algorithms in terms of efficiency and data utility.

Index Terms—natural equivalent class; data anonymization; privacy preserving; k -anonymity; clustering

I. INTRODUCTION

In collaborative environments, there are many kinds of data such as mobile phone data, Internet of things (IoT) data, network data, etc. To further exploit these data, data owners need to share them with data recipients or make them public. Since these data include sensitive information, publishers are usually appointed to process the data by adopting the anonymization techniques to protect the sensitive information, and guarantee data utility [1]. In order to prevent privacy disclosure, data anonymization technique often anonymizes quasi-identifiers (i.e. age, gender) that can be used to uniquely identify an individual once combined. After anonymization, quasi-identifier values of several records are equal, thus forming an equivalence class. In this way, an individual would not be easily identified because the attacker obtains only the equivalence class without additional knowledge [2].

State-of-the-art data anonymization approaches are mainly single-record-based algorithms. They often need to calculate the distance between each record and each cluster in the record allocation step, which can cause high computation complexity. In fact, we find that there is a kind of equivalence class naturally exists in the raw dataset, which we define as *natural equivalent class*. If we employ the natural equivalent class as a basic unit to be allocated, which means several records can be allocated at a time, it will reduce the computation

time. Besides, after checking several online datasets, we find some natural equivalent classes also meet the anonymization requirements. However, some clustering algorithms [3], [4] may separate natural equivalent classes during anonymization, which leads to additional information loss. Taking the widely used Adult Data Set as an example, we find that when choosing eight attributes as quasi-identifiers, about 53.51% records exist in the natural equivalent classes, each of which contains at least two records. If we could make use of these records contained in the natural equivalent classes, it would be more efficient for the dataset to have the k -anonymity property with less information loss.

In this paper, we propose the concept of natural equivalent class, and prove that utilizing natural equivalent class is able to effectively reduce computational complexity of the clustering algorithm as well as information loss. On such basis, we propose a novel data anonymization algorithm based on the natural equivalent class. It clusters data in unit of natural equivalent class, and realizes k -anonymity through generalization. Then we evaluate our algorithm based on the Adult Data Set. The results show that our approach outperforms the previous clustering-based anonymization algorithm in terms of efficiency and data utility.

The main contributions of our work are summarized as follows:

- We find that there naturally exists a kind of equivalence class in microdata, and define it as natural equivalent class.
- A novel data anonymization algorithm is proposed based on the natural equivalent class.
- We evaluate our algorithm with real world datasets, and compare it with state-of-the-art clustering-based anonymization algorithms.

The rest of this paper is organized as follows. We first discuss the related work in Section II and formalize the underlying concepts in Section III. In Section IV, we prove the effect of natural equivalent class theoretically and put forward the algorithm. Section V describes our experiments that demonstrate the advantage of our algorithm. Finally, we conclude this paper in Section VI.

II. RELATED WORK

Data anonymization techniques can be divided into two categories, namely privacy model and data anonymization algorithm. Researches on privacy model mainly focus on the analysis of privacy disclosure risk and the construction of constraint model in theory. Thus, it provides theoretical guidance for anonymization algorithms. Researches on data anonymization algorithms aim to design the efficient algorithms under the constraint of privacy model, for the purpose of minimizing information loss. Under particular model constraints, the algorithm with the lowest information loss is called the optimal anonymization algorithm. However, under the existing privacy model constraints, the optimal anonymization algorithm is NP-hard [5], [6], therefore, most anonymization algorithms are approximation algorithms.

In 2002, Sweeney and Samarati raised a problem that linking the quasi-identifiers of public datasets would cause privacy disclosure, and proposed the k -anonymity model [2]. The k -anonymity model requires that quasi-identifiers of each record are equal to at least $k - 1$ other records, ensuring that the dataset won't disclose users' privacy when facing linking attack. Meyerson *et al.* [5] proved that: when $k \geq 2$, k -anonymity is an NP-hard problem under minimum generalization and suppression. However, k -anonymity does not take the sensitive attributes into account. When most users have the same sensitive attributes, the attacker is very likely to deduce the users' sensitive information. Therefore, Raymond *et al.* [7] proposed (α, k) -anonymity model, which ensures the probability of sensitive attribute occurring in each equivalence class is less than α . To further improve the model safety, l -diversity [8], t -Closeness [9] and m -invariance [10] were proposed one after another. However, as each coin has two sides, the safer a privacy model is, the worse the data utility will be.

In the field of data anonymization algorithm, Sweeney [11] proposed a k -anonymity algorithm (MinGen) combining generalization and suppression. H. Park and K. Shim [12] presented an approximate algorithm with the $O(\log k)$ approximation ratio. LeFevre *et al.* [13] proposed a k -anonymity algorithm Incognito based on global generalization space pruning. This algorithm reduces the search space by dynamic planning, but it still leads to high information loss. Therefore, Xu *et al.* [3] presented a local generalization based on clustering, which can reduce the information loss by refining the generalization granularity. Byun *et al.* [4] proposed a clustering anonymization algorithm based on k -member, and later OKA [14] proved to be better. With the increase of data dimension, the information loss due to data generalization will increase rapidly [15]. To address this problem, Anatomy [16] was proposed to reduce the risk of privacy leakage, which weakens the relationship between quasi-identifiers and sensitive attributes through lossy decomposition. To further reduce the information loss, the algorithm ANGEL [17] was put forward.

All of the data anonymization algorithms above are based on each single record of the raw data, and they have not

utilized the natural equivalent class in the raw data. Some clustering algorithms, such as k -member [4] and OKA [14], even separate the natural equivalent class. Therefore, in this paper, we propose an anonymization algorithm based on the natural equivalent class.

III. BASIC CONCEPTS

In this paper, we focus on the relational dataset. All attributes are divided into four categories:

- **Personal Identifier:** the attributes which can identify the individuals, such as national identification number, name, E-mail and telephone number;
- **Quasi-Identifier (QID):** the attributes which can be joined together to uniquely identify the individuals, such as age, sex, postcode, etc.;
- **Sensitive Attribute (SA):** the attributes including the individual's sensitive information, such as medical information, personal income, home address, etc.;
- **Other attributes:** attributes that do not fall into the above three categories.

During data anonymization, personal identifier should be removed, QID needs to be properly anonymized, SA and other attributes should be fully preserved in most cases. In this paper, we define the published dataset as T , T^* represents the anonymized dataset, n means the number of records in T , the d represents the number of QIDs, record in T is defined as t , and v is the value of t .

Definition 1 (Equivalence Class, EC) [10]. An equivalence class includes records **with the same QID values.**

Definition 2 (k -anonymity) [1]. If each EC in dataset T^* contains at least k records, T^* meets k -anonymity.

To simplify discussion, we focus on k -anonymity in this paper. Our result can be extended to other privacy models.

Definition 3 (Generalization). Generalization is the procedure of replacing exact values with vague or range values during data anonymization.

In order to preserve the semantic meaning, we often generalize the QID step by step with pre-defined generalization hierarchies.

Definition 4 (Natural Equivalent Class, NEC). The equivalence class **which naturally exists in the raw dataset T** is NEC.

In NEC, all the records have the same value of QID. Particularly, if $|NEC| = 1$, it means there is only one record in this NEC. In practice, we let the value of QID be *key*, and each NEC correspond to a *key*. Then we utilize hash algorithm to cluster the records which have the same *key*. After doing that, we get the NEC, and the complexity of this step is $O(dn)$.

In data anonymization process, the decrease in the users' information is regarded as information loss. The value of information loss is calculated by the loss formula, defined as IL . The higher the information loss, the worse the data usability. In addition, as all the records have the same QID in NEC, we have $IL(NEC) = 0$.

IV. NEC-BASED DATA ANONYMIZATION ALGORITHM

After analyzing state-of-the-art anonymization algorithms, we find that utilizing NEC can efficiently reduce time complexity of clustering-based anonymization algorithms, which are designed to achieve higher data utility with higher time complexity. With the help of NEC, we can cluster records based on NEC rather than the single record.

A. Impact of NEC on the clustering-based algorithm

Most clustering-based anonymization algorithms, e.g., **OKA** and **k-member**, are centroid-based clustering algorithms. In these algorithms, they often have two main steps, one is **clustering building step** and the other is **record allocation step**. Besides, the IL metric, which is designed to measure the total information loss caused by anonymization, as often served as distance metric during clustering. Without loss of generality, we assume that the average size of NEC in T is m , $dist$ represents the distance metric, and C means a cluster.

According to [4] and [14], we define the relationship between $dist$ and IL :

$$dist(C, s) = IL(C \cup s) \quad (1)$$

where C represents a cluster or group, s is a set incorporated into C . According to the basic principles of data generalization [11], we have:

$$IL(C \cup s) - IL(C) \geq 0 \quad (2)$$

$$\frac{IL(C \cup s)}{|C| + |s|} - \frac{IL(C)}{|C|} \geq 0 \quad (3)$$

Theorem 1. For clustering-based anonymization algorithm, if NEC is a basic unit for clustering, its computation complexity is no higher than clustering based on record.

Proof. In clustering-based anonymization algorithms, the record allocation step has the maximum computation complexity. In this step, we need to calculate the distance between each record and each cluster, and allocate the record to the nearest cluster. The complexity of allocating one record is $O(n)$, and that of allocating n records is $O(n^2)$. According Definition 1 and 4, all records in the NEC have the same value of QID, so they have the same distance between the clusters. Therefore, we can allocate m records after calculating the distance once, and the complexity decreases from $O(n^2)$ to $O(n^2/m)$. In the worst condition, if $m = 1$, which means each NEC has only one record, the computation complexity is equal to $O(n^2)$ which is no higher than the original complexity.

Lemma 1. In clustering-based anonymization algorithm, splitting or merging the NEC may result in more information loss.

Proof. We assume $\exists nec$, and $|nec| \geq k$. In the clustering period, there are two cases:

- If nec is an independent cluster, we have its information loss:

$$IL(nec) + \sum_{i=1}^x IL(C_i) \quad (4)$$

- If nec is divided into a disjoint set $\{s_1, \dots, s_x\}$, then add s_i into different cluster $\{C_1, \dots, C_x\}$, s_i can be empty and $|nec| = \sum_{i=1}^x |s_i|$. We have its information loss:

$$\sum_{i=1}^x IL(C_i \cup s_i) \quad (5)$$

To prove the Lemma 1 equals to prove that Formula 5 is no less than Formula 4. So we let Formula 5 minus Formula 4:

$$\sum_{i=1}^x (IL(C_i \cup s_i) - IL(C_i)) - IL(nec) \quad (6)$$

According to the definition of information loss, we have $IL(nec) = 0$. Meanwhile, according to the fundamental property of generalization and Formula 2, we have $\sum_{i=1}^x (IL(C_i \cup s_i) - IL(C_i)) \geq 0$, so Formula 6 is no less than 0. In the other word, Formula 5 is no less than Formula 4. Note that formula 6 is equal to 0, if and only if all records in s_i and in C_i are totally the same on QIDs. That is quite hard to satisfy in raw dataset.

Lemma 2. In the record allocation step of clustering algorithm, if all clusters have the same number of records, we can achieve the minimum information loss by allocating the NEC as the basic unit.

Proof. We assume $\exists nec$, and $|nec| < k$. The distance between nec and each cluster meets the following relationship: for any i and j , if $i < j$, we have $dist(C_i, nec) \leq dist(C_j, nec)$, and it is equivalent to $IL(C_i \cup nec) \leq IL(C_j \cup nec)$. Since the number of records in all clusters is the same, we have $|C_i| + |nec| = w$, and $w \geq 0$. In the clustering period, there are two ways to process nec :

- The nec is divided into the nearest cluster C_1 , and its information loss is:

$$IL(C_1 \cup nec) + \sum_{i=1}^x IL(C_i) \quad (7)$$

- The nec is divided into disjoint set $\{s_1, \dots, s_x\}$, and respectively join in different clusters $\{C_1, \dots, C_x\}$. In addition, s_i can be empty, and $|nec| = \sum_{i=1}^x |s_i|$. Its information loss is:

$$\sum_{i=1}^x IL(C_i \cup s_i) \quad (8)$$

To prove Lemma 2 correctness equals to prove that Formula 8 is equal to or greater than Formula 7. So, we let Formula 8 minus Formula 7:

$$\sum_{i=2}^x (IL(C_i \cup s_i) - IL(C_i)) + IL(C_1 \cup s_1) - IL(C_1 \cup nec) \quad (9)$$

As the QIDs of nec and s_i have the same value, the generalization result should be the same after they are merged with C_i , therefore, we have:

$$\frac{IL(C_i \cup s_i)}{|C_i| + |s_i|} = \frac{IL(C_i \cup nec)}{|C_i| + |nec|} \quad (10)$$

Then we substitute Formula 10 into Formula 9, and simplify the $IL(C_i)$ with Formula 3:

$$\sum_{i=2}^x (IL(C_i \cup s_i) - IL(C_i)) + IL(C_1 \cup s_1) - IL(C_1 \cup nec) \geq \sum_{i=2}^x \left(\frac{|s_i| IL(C_i \cup s_i)}{|C_i| + |s_i|} - \frac{|s_i| IL(C_1 \cup s_1)}{|C_1| + |s_1|} \right) \quad (11)$$

Then we substitute Formula 10 into Formula 11, and substitute $|C_i| + |nec| = w$, we have:

$$\sum_{i=2}^x \frac{|s_i|}{w} (IL(C_i \cup nec) - IL(C_1 \cup nec)) \quad (12)$$

According to our initial assumption, when $i \geq 2$, we have $IL(C_i \cup nec) \geq IL(C_1 \cup nec)$. Therefore, Formula 12 and Formula 9 are definitely not less than 0.

Theorem 2. In certain conditions, clustering with NEC can achieve minimum information loss in clustering building step and record allocation step.

According to Lemma 1, building cluster based on NEC can decrease the information loss; according to Lemma 2, when the amount of clustering is the same, NEC-based clustering can decrease the information loss during the record allocation step.

The theoretical results above show that, by utilizing NEC in the clustering-based anonymization algorithm, we can improve both data utility and anonymization efficiency.

B. Optimization of clustering algorithm by NEC

On the basis of the foregoing theoretical demonstration and data anonymization algorithms i.e., OKA and k -member, we design two new algorithms named NEC_based_OKA and NEC_based_K-member, the basic process is shown below:

Algorithm 1 NEC_based_OKA/K-member

Input: k , the original dataset T

Output: The anonymization dataset T^*

- 1: Find all NECs in the T to build the set, named NEC_SET
- 2: **for** each NEC in NEC_SET **do**
- 3: **if** $|NEC| \geq k$ **then**
- 4: Cluster_SET $\leftarrow$ NEC
- 5: NEC_SET = NEC_SET - NEC
- 6: **end if**
- 7: **end for**
- 8: OKA or k -member cluster the left NECs in NEC_SET, Cluster_SET $\leftarrow$ cluster
- 9: **If** $\exists$ NECs $\notin$ Cluster_SET, add them into the nearest cluster
- 10: **Generalize the cluster in Cluster_SET, and add them into T^***
- 11: **return** T^*

There are 5 steps in the NEC-based anonymization algorithm:

- 1) Go through all records in T , build NEC_SET according to QID. With hash algorithm, the computation complexity is $O(dn)$ (d is the number of QIDs in T).

TABLE I: Adult dataset

Attribute Type	Attribute Name	Cardinality	Data Type
	Age	74	numeric
QID	Gender	2	categorical(2)
	Education Level	17	numeric
	Marital Status	7	categorical(3)
	Race	5	categorical(2)
	Work Class	8	categorical(3)
	Native Country	41	categorical(2)
	Occupation	14	categorical(2)
SA	Salary Class	2	categorical

- 2) Take each NEC which is larger than k as an independent cluster, add it into the Cluster_SET, and the complexity is $O(n/m)$ (m is the average number of records in NEC).
- 3) Utilize OKA/ k -member to cluster the left NECs in NEC_SET, and the complexity is $O(n^2/m)$.
- 4) For NEC that cannot be clustered, calculate the distance between them and the cluster in Cluster_SET, add them into the nearest cluster, and the complexity is $O(n/m)$.
- 5) Generalize the cluster in Cluster_SET, make the value of QID the same, and the complexity is $O(n/m)$.

In this algorithm, OKA and k -member are used in the clustering period. Since OKA and k -member are easier to implement and work better compared with other state-of-the-art clustering algorithms. Therefore, we choose them as a subalgorithm to cluster NECs. The computation complexity of the whole algorithm is $O(n^2/m)$, and the value of m is influenced by the dataset scale and data dimension.

V. EXPERIMENTS EVALUATION

A. Experimental Setup

In our experiments, we choose Adult Data Set which is widely used in the data anonymization research [18] as the test dataset. This dataset is part of United States' 1996 demographics. We choose 9 attributes from dataset. According to [13], we consider Age, Gender, Race, Marital Status, Education Level, Native Country, Work Class, and Occupation as the QID, Salary Class as sensitive attribute. The characteristics of the dataset are summarized in Table I. Among the QIDs, Marital Status and Work Class are the multi-hierarchical structures, and the other attributes have two generalization hierarchies.

In the data anonymization, the utility of the anonymized dataset is measured by the degree of information loss. The higher the degree of information loss, the worse the data utility. To efficiently measure the information loss of the dataset, many measurement formulas are proposed, such as Classification Metric (CM) [19], Discernibility Metric (DM) [20], and Normalized Certainty Penalty (NCP) [13], Global Certainty Penalty (GCP) [21]. NCP and GCP, which can calculate the distance, are the most commonly used ones. According to [21], GCP is one type of normalization of NCP. The definition of numeric data's NCP is:

$$NCP(v) = \frac{v_{max} - v_{min}}{A_{max} - A_{min}} \quad (13)$$

In Formula 13, v represents the value of the A attribute, v_{max} and A_{max} represent the maximum of v and A respectively.

Similarly, v_{min} and A_{min} represent the minimum of v and A . The definition of categorical data's NCP is:

$$NCP(v) = \begin{cases} 0 & |v| = 1 \\ |v|/|A| & otherwise \end{cases} \quad (14)$$

In Formula 14, $|A|$ represents the number of leaf nodes in generalization hierarchy of the A attribute, and $|v|$ means the number of leaf nodes covered by v . According to Formula 13 and 14, if one value is not generalized, $NCP = 0$; if $NCP = 1$, it means this value is suppressed. NCP can indicate information loss amount of a dataset through simple accumulation, while it is not user friendly. Here, we use GCP which represents the ratio of the information loss of a dataset:

$$GCP(T) = \frac{\sum_{i=1}^n NCP(t_i)}{dn} \quad (15)$$

The value range of $GCP(T)$ is $[0, 1]$, which we often convert it into percentage. For example, if $GCP(T) = 20\%$, it means the dataset lost 20% information during anonymization.

To make a fair comparison, we adopt NCP to measure the distance, and utilize GCP to measure the information loss of the anonymization algorithm. In addition, the average runtime demonstrates the anonymization efficiency of the algorithm. The hardware environment of the experiments: Intel Xeon E5-2620v2 CPU, 60G RAM, OS Redhat 7. NEC_based_OKA, NEC_based_k-member, OKA and k-member algorithms are implemented in Python.

B. The experimental results and analysis

During data anonymization, the value of k , the dimensionality of QID d , and the size of the dataset are three main factors that affect the final result. In this paper, we evaluate our algorithms by changing these three factors. Since any randomly selected sub-dataset in a designated size is different, we randomly choose 10 sub-datasets and take the average result of these 10 sub-datasets as the final result.

1) *Adult dataset records distribution*: In the dataset, the ratio and distribution of NEC will change with the dimension of QID and the size of the dataset. We calculate the ratio of records owned by NEC with at least k records, for these records are crucial to improve the anonymization effect.

Fig. 1 shows the relationship among k , d and the NEC records distribution of the adult dataset. When we choose eight QIDs from the dataset, there are 53.51% records in the NECs with $|NEC| \geq 2$. With the decrease in d , the number of records satisfying k -anonymity increases. The reason is that, the decrease in d will increase the data density, raising the overlap probability of records. Similarly, we also find that as the size of dataset grows, the number of NEC increases.

2) *Data utility (information loss)*: In Fig. 2, it shows that with the increase of k , the information loss of four algorithms increases accordingly. The larger k is, the more records in one cluster there are, and the higher the information loss is. It can also be seen that under the same conditions, the information loss of NEC_based_OKA is less than that of OKA; the information loss is nearly equal between NEC_based_k-member

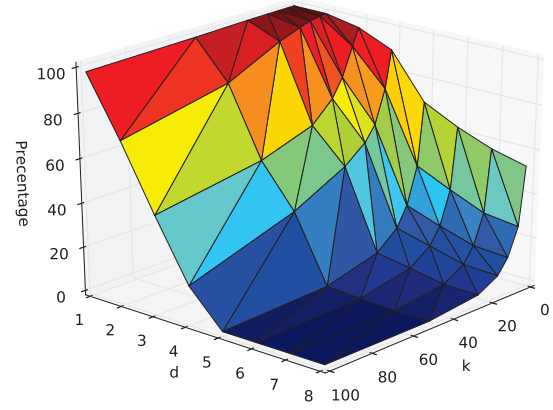


Fig. 1: NEC Records Distributions vs. d and k

and k -member when k is small, however, when k grows, the NEC_based_k-member performs better than k -member in information loss.

As Fig. 3 shows, with the increase of d , the information loss of four algorithms increases slowly. The larger the d is, the sparser the data is, and the harder it is to realize k -anonymity. When the value of d changes, it can be seen that the NEC_based_OKA performs better than OKA, and the NEC_based_k-member is nearly equal to k -member in information loss.

From Fig. 4, we find that with the growing of the size of dataset, the information loss of four algorithms decreases slowly. As the size of dataset grows, it is more likely to have the k -anonymity property. We also find that, the NEC-based algorithms achieve less information loss than the previous algorithms.

3) *Efficiency*: In Fig. 5, with the increase of k , the execution time of NEC_based_k-member and k -member will be longer and longer, because the larger k requires more times of traversing the dataset, thus reducing the anonymization efficiency. On the country, the NEC_based_OKA and OKA algorithms need to build fewer clusters with the increase of k , contributing to shorter execution time. In all values of k , the running time of NEC_based_k-member is much shorter than k -member, especially, when $k=100$, the efficiency of NEC_based_k-member is 3 times that of k -member. The running time of NEC_based_OKA is slightly shorter than that of OKA.

As shown in Fig. 6, the increasing d results in more running time of four algorithms, for the high dimension increases the overhead of distance computing. We can also find that, the NEC-based algorithms are more efficient than previous algorithms.

Fig. 7 demonstrates that the computing efficiency decreases with the increase of dataset size. It's easy to understand that the more records there are, the longer computing time it takes. It is also shown that, with the size growing, the advantages of NEC-based algorithms will be more obvious.

After analyzing our experimental results, we find that all NEC-based algorithms have the same trend with the previous

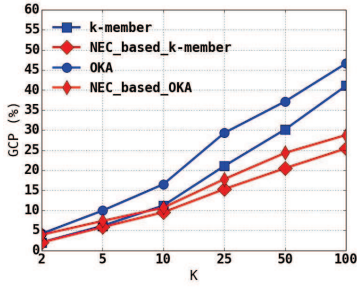


Fig. 2: GCP vs. k

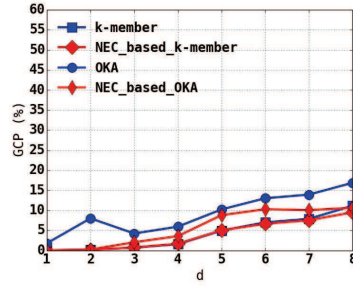


Fig. 3: GCP vs. d

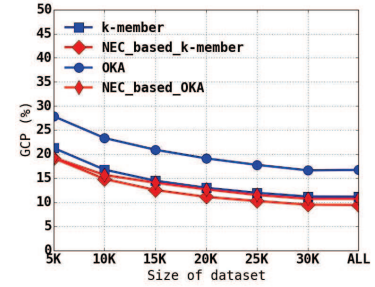


Fig. 4: GCP vs. size

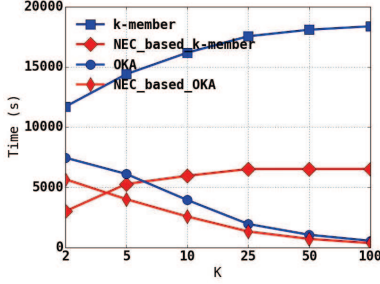


Fig. 5: Time vs. k

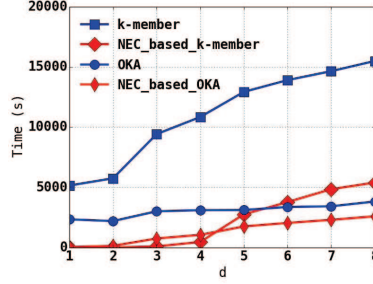


Fig. 6: Time vs. d

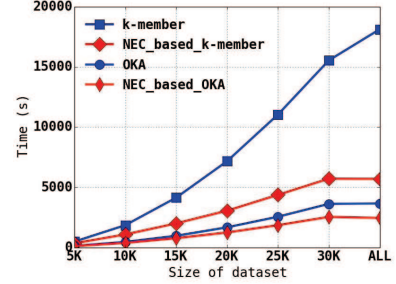


Fig. 7: Time vs. size

algorithms. And the NEC-based algorithms perform better than the unimproved algorithms in both data utility and anonymization efficiency.

VI. CONCLUSION

In this paper, we propose a new concept named NEC which refers to an equivalence class naturally existing in the raw dataset, and design the algorithms accordingly. These algorithms cluster the records on the basis of NEC. Compared with the previous clustering algorithms on the real-world datasets, our algorithm reduces the information loss and promotes the anonymization efficiency. In the future, we are going to extend NEC to other anonymization algorithms and perform more experiments to demonstrate the effects of NEC on other datasets and algorithms.

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